

Transcendental Algebraic Geometry

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Abstract

Contents

1	The GAGA Principle	3
1.1	The Analytification Functor	3
1.2	Algebraic de Rham Cohomology	4
1.3	The Hodge de Rham Spectral Sequence	7
1.4	Comparing Cohomology Theories	8
2	Compact Kahler Manifolds	9
2.1	Equivalence to Smooth Projective Varieties	9
2.2	Hodge Classes	9
2.3	The Lefschetz Theorems	10
2.4	Projective Curves	10
3	Mixed Hodge Structures	12
4	Periods and Period Maps	13

1 The GAGA Principle

1.1 The Analytification Functor

Recall that the category of schemes and the category of analytic spaces are both full subcategories of locally ringed spaces.

Definition 1.1.1 (The Analytification of a Scheme)

Let X be a scheme locally of finite type over $\mathbb{C}$. Define the analytification of X to be the unique analytic space X^{an} together with a morphism of locally ringed space $\alpha : X^{\text{an}} \rightarrow X$ satisfying the following universal property: For any analytic space Y and any morphism of locally ringed spaces $f : Y \rightarrow X$, there exists a unique morphism of analytic spaces $g : Y \rightarrow X^{\text{an}}$ such that the following diagram commutes:

$$\begin{array}{ccc} & Y & \\ \exists! g \swarrow & \downarrow f & \\ X^{\text{an}} & \xrightarrow{\alpha} & X \end{array}$$

Recall that the Yoneda embedding $h : \mathbf{LRS} \rightarrow \mathbf{Set}^{\mathbf{LRS}}$ is the functor sending a locally ringed space X to the functor $h_X : \mathbf{LRS} \rightarrow \mathbf{Set}$ defined by $h_X(Y) = \text{Hom}_{\mathbf{LRS}}(Y, X)$. Then the universal property of the analytification means that the functor h_X restricted to analytic spaces $\mathbf{An}$ is representable.

One thinks of the morphism $X^{\text{an}} \rightarrow X$ as doing two things. It is an enlargement of the standard topology of X^{an} to the Zariski topology of X , and it introduces generic points outside of $X(\mathbb{C})$ (Indeed we will see that X^{an} has the same underlying set as $X(\mathbb{C})$).

Theorem 1.1.2

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the analytification of X exists and is unique up to unique isomorphism.

Definition 1.1.3 (The Analytification Functor)

Define the analytification functor

$$(-)^{\text{an}} : \mathbf{Sch}_{\mathbb{C}} \rightarrow \mathbf{An}$$

as follows.

- For each scheme X locally of finite type over $\mathbb{C}$, we have X^{an} is the analytification of X .
- For a morphism of schemes $f : X \rightarrow Y$ over $\mathbb{C}$, the universal property of Y^{an} induces a morphism $X^{\text{an}} \rightarrow Y^{\text{an}}$. Precomposing with the canonical morphism $X^{\text{an}} \rightarrow Y^{\text{an}}$ gives the desired map $X^{\text{an}} \rightarrow Y^{\text{an}}$.

Proposition 1.1.4

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the underlying set of X^{an} is given by $X(\mathbb{C})$.

Proposition 1.1.5

Let X be a scheme locally of finite type over $\mathbb{C}$. Then the following are true.

- X is separated if and only if X^{an} is Hausdorff.
- X is connected if and only if X^{an} is connected.
- X is smooth if and only if X^{an} is a complex manifold.
- X is proper if and only if X^{an} is compact.

Proposition 1.1.6

Let X be a variety over $\mathbb{C}$. Then the following are true.

- If X is irreducible, then X^{an} is connected.
- If X is smooth and affine, then X^{an} is a Stein manifold.
- If X is smooth and projective, then X^{an} is Kahler.

Theorem 1.1.7 (GAGA)

Let X be a proper scheme of finite type over $\mathbb{C}$. Then the following are true.

- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Then $\mathcal{F}^{\text{an}}$ is a sheaf of $\mathcal{O}_{X^{\text{an}}}$ -module.
- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. There is a natural isomorphism

$$H^i(X, \mathcal{F}) \cong H^i(X^{\text{an}}, \mathcal{F}^{\text{an}})$$

- The analytification functor induces an equivalence of categories

$$\mathbf{Coh}_{\mathcal{O}_X} \cong \mathbf{Coh}_{\mathcal{O}_{X^{\text{an}}}}$$

- Let $Y \subseteq X^{\text{an}}$ be a closed complex analytic subspace. Then there exists a closed subvariety Z of X such that $Z^{\text{an}} = Y$.

Corollary 1.1.8 (Chow's Theorem)

Let X be a closed complex analytic subspace of $\mathbb{CP}^n$. Then X is a projective variety over $\mathbb{C}$.

The important point is that: the theory of smooth projective complex varieties are subsumed by the theory of Kahler manifolds. Divisors and line bundles correspond bijectively thanks to Chow's theorem.

1.2 Algebraic de Rham Cohomology**Definition 1.2.1 (Algebraic de Rham Cohomology)**

Let S be a scheme. Let X be a scheme over S . Define the n th algebraic de Rham cohomology group by

$$H_{\text{dR}}^n(X/S) = \mathbb{H}^n(X; \Omega_{X/S}^\bullet)$$

Lemma 1.2.2

Let S be a scheme. Let X be a scheme over S . Then $\text{Tot}^\oplus \left(C^\bullet(X, \mathcal{U}, \Omega_{X/S}^\bullet) \right)$ is an acyclic resolution of $\Omega_{X/S}^\bullet$.

Thus one way to compute the de Rham cohomology would be

$$H_{\text{dR}}^n(X/S) \cong H^n(\text{Tot}^\oplus(C^\bullet(X, \mathcal{U}, \Omega_{X/S}^\bullet)))$$

Example 1.2.3

We have

$$H_{\text{dR}}^n(\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}) \cong \begin{cases} \mathbb{C} & \text{if } k = 0, 2 \\ 0 & \text{otherwise} \end{cases}$$

Proof

Consider the canonical affine cover $\mathcal{U} = \{U_0, U_1\}$ of $\mathbb{P}_{\mathbb{C}}^1$. We have

$$C^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{U}, \Omega_{\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}}^k) = \Omega_{\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}}^k(U_0) \times \Omega_{\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}}^k(U_1) \cong \begin{cases} \mathbb{C}[x]dx \oplus \mathbb{C}[y]dy & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$C^1(\mathbb{P}_{\mathbb{C}}^1, \mathcal{U}, \Omega_{\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}}^k) = \Omega_{\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}}^k(U_0 \cap U_1) \cong \begin{cases} \mathbb{C}[x, x^{-1}]dx & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

Also we have

$$C^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{U}, \mathcal{O}_{\mathbb{P}_{\mathbb{C}}^1}) \cong \mathbb{C}[x] \oplus \mathbb{C}[y] \quad \text{and} \quad C^1(\mathbb{P}_{\mathbb{C}}^1, \mathcal{U}, \mathcal{O}_{\mathbb{P}_{\mathbb{C}}^1}) \cong \mathbb{C}[x, x^{-1}]$$

so all the data assembles into the diagram

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \uparrow & & \uparrow & & \\
0 & \longrightarrow & \mathbb{C}[x]dx \oplus \mathbb{C}[y]dy & \xrightarrow{(f dx, g dy) \mapsto (-x^{-2}g(x^{-1}) - f(x))dx} & \mathbb{C}[x, x^{-1}]dx & \longrightarrow & 0 \\
& & \uparrow & & \uparrow & & \\
& & (f, g) \mapsto (df, dg) & & f \mapsto df & & \\
0 & \longrightarrow & \mathbb{C}[x] \oplus \mathbb{C}[y] & \xrightarrow{(f, g) \mapsto g(x^{-1}) - f(x)} & \mathbb{C}[x, x^{-1}] & \longrightarrow & 0 \\
& & \uparrow & & \uparrow & & \\
& & 0 & & 0 & &
\end{array}$$

Then the total complex is given by

$$0 \longrightarrow \mathbb{C}[x] \oplus \mathbb{C}[y] \xrightarrow{(f, g) \mapsto (f(x) - g(x^{-1}), df, dg)} \mathbb{C}[x]dx \oplus \mathbb{C}[y]dy \oplus \mathbb{C}[x, x^{-1}]dx \xrightarrow{g dx, h dy, f \mapsto df + (-x^{-2}h(x^{-1}) - g(x))dx} \mathbb{C}[x, x^{-1}]dx \longrightarrow 0$$

Then we have $H_{\text{dR}}^0(\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}) \cong \mathbb{C}$ and $H_{\text{dR}}^1(\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}) = \frac{\mathbb{C}[x, x^{-1}]dx}{\mathbb{C}[x, x^{-1}]dx} = 0$ and $H_{\text{dR}}^2(\mathbb{P}_{\mathbb{C}}^1/\mathbb{C}) \cong \mathbb{C}\langle x^{-1}dx \rangle \cong \mathbb{C}$ ■

Lemma 1.2.4

Let k be a field of characteristic 0. Let X be a smooth affine variety over k . Then we have

$$H_{\text{dR}}^n(X) = H^n(\Gamma(X, \Omega_{X/k}^\bullet))$$

for all $n \in \mathbb{N}$.

Example 1.2.5

Let $X = \text{Spec}(\mathbb{C}[x, x^{-1}])$. Then we have

$$H_{\text{dR}}^n(X/\mathbb{C}) \cong \begin{cases} \mathbb{C} & \text{if } n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof

Since X is smooth and affine over $\mathbb{C}$, we just have to take global sections. The relevant chain complex is then $0 \rightarrow \mathcal{O}_X(X) \rightarrow \Omega_{X/\mathbb{C}}^1(X) \rightarrow 0$. Since X is affine, we have that $\mathcal{O}_X(X) = \mathbb{C}[x, x^{-1}]$. To compute $\Omega_{X/\mathbb{C}}^1(X)$, notice that $\Omega_{X/\mathbb{C}}^1(X) = \Omega_{\mathbb{C}[x, x^{-1}]/\mathbb{C}}^1$. Then we have

$$\Omega_{\mathbb{C}[x, x^{-1}]/\mathbb{C}}^1 \cong \left(\Omega_{\mathbb{C}[x]/\mathbb{C}}^1 \right)_x \cong (\mathbb{C}[x]dx)_x \cong \mathbb{C}[x, x^{-1}]dx$$

The differential $\mathcal{O}_X(X) \rightarrow \Omega_{X/\mathbb{C}}^1(X)$ is then just the total differential $f \mapsto df$. Clearly we have $H_{\text{dR}}^0(X/\mathbb{C}) = \ker(f \mapsto df) = \mathbb{C}$, and we have

$$H_{\text{dR}}^1(X/\mathbb{C}) = \frac{\mathbb{C}[x, x^{-1}]dx}{\mathbb{C}\langle x^n \mid n \in \mathbb{Z} \setminus \{-1\} \rangle} \cong \mathbb{C}\langle x^{-1}dx \rangle \cong \mathbb{C}$$

Proposition 1.2.6

Let S be a scheme. Let X be a scheme over S . Then the following are true.

- $H_{\text{dR}}^n(X/S)$ is a $H_{\text{dR}}^0(X/S)$ -module.
- The map $H_{\text{dR}}^i(X/S) \times H_{\text{dR}}^j(X/S) \rightarrow H_{\text{dR}}^{i+j}(X/S)$ induced by the wedge of differential forms is a $H_{\text{dR}}^0(X/S)$ -linear.

Definition 1.2.7 (The de Rham Cohomology Ring)

Let S be a scheme. Let X be a scheme over S . Define the de Rham cohomology ring of X to be

$$H_{\text{dR}}^*(X/S) = \bigoplus_{i=0}^{\infty} H_{\text{dR}}^i(X/S)$$

together with ring structure given by wedge of differential forms.

Lemma 1.2.8

Let S be a scheme. Let X be a scheme over S . Then $H_{\text{dR}}^*(X/S)$ is a graded commutative $H_{\text{dR}}^0(X/S)$ -algebra.

Example 1.2.9

Let k be a field of characteristic 0. Let $R = k[x_1, \dots, x_n]$ be the polynomial ring over k . Then $\Gamma(X, \Omega_{\text{Spec}(R)/k}^1) = \Omega_{R/k}^1 = \bigoplus_{i=1}^n R dx_i$. Then we have

$$\Gamma(X, \Omega_{\text{Spec}(R)/k}^j) = \bigoplus_{1 \leq i_1 < \dots < i_j \leq n} R dx_{i_1} \wedge \dots \wedge dx_{i_j}$$

Then we have

$$H_{\text{dR}}^i(R/k) = \begin{cases} k & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

TBR:

Example 1.2.10

Let k be a field of characteristic $p > 0$. Then we have

$$\dim_k(H^1(\Gamma(\text{Spec}(k[t]), \Omega_{\text{Spec}(k[t])/k}^1))) = \infty$$

Indeed, we have

$$H^1(\Gamma(\text{Spec}(k[t]), \Omega_{\text{Spec}(k[t])/k}^1)) = \frac{k[t]}{\langle f'(t) \mid f \in k[t] \rangle} = \frac{k[t]}{\bigoplus_{i=1}^{\infty} t^{i-1} dt} \cong \bigoplus_{i=1}^{\infty} t^{i-1} dt$$

Example 1.2.11

$$\Omega_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}}^1 \cong (\mathbb{Z}/2\mathbb{Z})d\sqrt{2}. \quad H^1(\Gamma(\text{Spec}(\mathbb{Z}\sqrt{2}), \Omega_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}}^1)) \neq 0$$

Example 1.2.12

Let k be a field of characteristic $p \neq 2$. Let $f \in k[x]$ be separable. Let $R = \frac{k[x,y]}{y^2 - f(x)}$. Then we have

$$\Omega_{R/k}^1 \cong \frac{\Omega_{k[x,y]/k}^1 \otimes_{k[x,y]} k}{(2ydy - f'(x)dx)} \cong \frac{Rdx \oplus Rdy}{R(2ydy - f'(x)dx)}$$

Using the quotient relation, we may write $dy = \frac{f'(x)dx}{2y}$ in the module. Thus dx/y is a generator of the module.

Since f is separable, there exists $g, h \in k[x]$ such that $gf + hf' = 1$. Let $\omega = aydx + 2bdy \in \Omega_{R/k}^1$. Then we have $y\omega = dx$. Hence y divides dx and $\omega = dx/y$.

Now let $z = a(x) + b(x)y \in R$ be arbitrary. Then we have

$$dz = (a'(x) + b'(x)y)dx + h(x)dy = (ya'(x) + b'(x)f(x) + \frac{1}{2}b(x)f'(x))\omega$$

Thus every element of $\Omega_{R/k}^1$ is of the form $q(x)\omega$ for some $q(x) \in k[x]$. If $\deg(q) \geq \deg(f)$, then

there exists $p \in k[x]$ such that $\deg(p) > \deg(q)$ such that

$$[p(x)\omega] = [q(x)\omega]$$

in $H_{\text{dR}}^1(R/k)$.

Lemma 1.2.13

Let k be a field of characteristic 0. Let X be a smooth variety over k . Then the following are true.

- $H_{\text{dR}}^i(X) = 0$ for all $i > 2 \dim(X)$.
- If X is affine, then $H_{\text{dR}}^i(X) = 0$ for all $i > \dim(X)$.

Proposition 1.2.14

1.3 The Hodge de Rham Spectral Sequence

Definition 1.3.1 (The Hodge Filtration)

Let S be a scheme. Let X be a scheme over S . Define the Hodge filtration $F^\bullet \Omega_{X/S}^\bullet$ of X to be

$$F^i \Omega_{X/S}^j = \begin{cases} \Omega_{X/S}^j & \text{if } j \geq i \\ 0 & \text{otherwise} \end{cases}$$

Thus for each i , the collection of complexes

$$F^i \Omega_{X/S}^\bullet = (0 \rightarrow \Omega_{X/S}^i \rightarrow \Omega_{X/S}^{i+1} \rightarrow \cdots)$$

form the filtration for $\Omega_{X/S}^\bullet$.

Lemma 1.3.2

Let S be a scheme. Let X be a scheme over S . Then $F^\bullet \Omega_{X/S}^\bullet$ is an exhaustive filtration.

Definition 1.3.3 (The Hodge de Rham Spectral Sequence)

Let S be a scheme. Let X be a scheme over S . Define the Hodge de Rham spectral sequence to be the spectral sequence associated to the Hodge filtration $F^\bullet \Omega_{X/S}^\bullet$.

The E^0 page is given by

$$E_0^{p,q} = \frac{F^p \Omega_{X/S}^{p+q}}{F^{p+1} \Omega_{X/S}^{p+q}} = \frac{\Omega_{X/S}^q}{\Omega_{X/S}^{q-1}}$$

Lemma 1.3.4

Let S be a scheme. Let X be a scheme over S . Then we have

$$E_1^{p,q} = H^q(X, \Omega_{X/S}^p) \Rightarrow H_{\text{dR}}^{p+q}(X/S)$$

Theorem 1.3.5 (Deligne)

If $S = \text{Spec}(k)$, k char 0, $X \rightarrow S$ smooth and proper. Then the Hodge de Rham spectral sequence degenerates at the E_1 page (all differentials in page 1 is 0).

When you have a 2-step filtration: $0 \rightarrow F^1 C^\bullet \rightarrow C^\bullet \rightarrow C^\bullet / F^1 C^\bullet \rightarrow 0$, the spectral sequence is the long exact sequence in cohomology.

1.4 Comparing Cohomology Theories

Theorem 1.4.1 (Grothendieck)

Let X be a smooth variety over $\mathbb{C}$. Then there is a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{Holo}}^n(X^{\mathrm{an}}) \cong H_{\mathrm{dR}}^n(X^{\mathrm{an}})$$

induced by the canonical map $\alpha : X^{\mathrm{an}} \rightarrow X$.

Corollary 1.4.2

Let X be a smooth variety over $\mathbb{C}$. Then the above isomorphism and de Rham's theorem induces a natural isomorphism

$$H_{\mathrm{dR}}^n(X) \cong H_{\mathrm{sing}}^n(X^{\mathrm{an}}; \mathbb{C})$$

Theorem 1.4.3 (The Period Isomorphism)

Let k be a subfield of $\mathbb{C}$. Let X be a smooth variety over k . Then there is a natural isomorphism

$$H_{\mathrm{dR}}^*(X) \otimes_k \mathbb{C} \cong H_{\mathrm{sing}}^*(X^{\mathrm{an}}; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$$

Map of Chow ring to singular cohomology.

2 Compact Kahler Manifolds

2.1 Equivalence to Smooth Projective Varieties

Proposition 2.1.1

Let X be a smooth projective variety over $\mathbb{C}$. Let $\mathcal{L}$ be a line bundle over X . Then $\mathcal{L}$ is ample if and only if $\mathcal{L}^{\text{an}}$ is positive.

Theorem 2.1.2 (Kodaira Embedding)

Let X be a compact complex manifold. Then the following are equivalent.

- X admits a holomorphic embedding into $\mathbb{CP}^n$.
- X admits an ample line bundle.
- X admits a positive line bundle.
- X admits a Kahler metric ω such that $[\omega] \in H^{1,1}(X, \mathbb{Z})$.
- There exists a projective variety Z over $\mathbb{C}$ such that $Z^{\text{an}} = X$.

Under the GAGA principle, we establish a dictionary.

Smooth projective notion	Compact Kähler notion	Comparison theorem
$\text{Pic}(X)$	$\text{Pic}(X^{\text{an}})$	GAGA (Serre)
$\text{Coh}_{\mathcal{O}_X}$	$\text{Coh}_{\mathcal{O}_{X^{\text{an}}}}$	GAGA
Algebraic subvarieties	Closed analytic subvarieties	Chow's theorem
$\text{Hom}_{\mathbf{Var}}(X, Y)$	$\text{Hom}_{\text{hol}}(X^{\text{an}}, Y^{\text{an}})$	GAGA + Chow
$H_{dR}^k(X/\mathbb{C})$	$H_{dR}^k(X^{\text{an}}, \mathbb{C}) = H_{\text{hol}}^k(X^{\text{an}})$	Grothendieck comparison
Hodge filtration	Hodge filtration	Grothendieck comparison
Ample line bundle	Positive line bundles	
Closed embeddings to projective space	Holomorphic embeddings to projective space	

In particular we have an equivalence of categories

Smooth Projective Varieties $\simeq$ Compact Kahler Manifolds with a choice of Kahler metric $[\omega] \in H^{1,1}(X; \mathbb{Z})$

Theorem 2.1.3 (Kodaira-Nakano Vanishing Theorem)

Let X be a compact Kahler manifold. Let $p : L \rightarrow X$ be a positive line bundle. Then we have

$$H^q(X, \Omega^p(L)) = 0$$

for $p + q > \dim(X)$.

2.2 Hodge Classes

Definition 2.2.1 (Hodge Classes)

Let X be a smooth projective variety. Define the p th Hodge classes of X to be

$$\text{Hdg}^p(X) = H^{2p}(X; \mathbb{Z}) \cap H^{p,p}(X)$$

The Lefschetz $(1, 1)$ -theorem says that chern classes of a compact Kahler manifold X corresponds to Hodge class $\text{Hdg}^1(X)$.

Definition 2.2.2 (The Algebraic Cycles of Smooth Subvarieties)

Let X be a smooth projective variety. Let $Z \subseteq X$ be a smooth closed, irreducible complex analytic

subvariety of Z . Define the Hodge class of Z by

$$[Z] = D^{-1}(\iota_*[Z]_Z) \in H^{\dim_{\mathbb{R}}(X) - \dim_{\mathbb{R}}(Z)}(X; \mathbb{Z})$$

where $\iota_* : H^{\dim_{\mathbb{R}}(Z)}(Z; \mathbb{Z}) \rightarrow H^{\dim_{\mathbb{R}}(Z)}(X; \mathbb{Z})$ is induced by inclusion $\iota : Z \hookrightarrow X$ and $D : H^{\dim_{\mathbb{R}}(X) - \dim_{\mathbb{R}}(Z)}(X; \mathbb{Z}) \rightarrow H^{\dim_{\mathbb{R}}(Z)}(X; \mathbb{Z})$ is the Poincaré duality map.

Lemma 2.2.3

Let X be a smooth projective variety. Let $Z \subseteq X$ be a smooth closed, irreducible complex analytic subvariety of Z . Then $[Z] \in \text{Hdg}^{\dim_{\mathbb{C}}(X) - \dim_{\mathbb{C}}(Z)}(X)$.

Definition 2.2.4 (The Algebraic Cycles of General Subvarieties)

Lemma 2.2.5

Let X be a smooth projective variety. Let $Z \subseteq X$ be a closed, irreducible complex analytic subvariety of Z . Then $[Z] \in \text{Hdg}^{\dim_{\mathbb{C}}(X) - \dim_{\mathbb{C}}(Z)}(X)$.

Definition 2.2.6 (The Cycle Homomorphism)

Let X be a smooth projective variety. Define the cycle homomorphism to be the map $\text{cl} : \text{CH}^p(X) \rightarrow \text{Hdg}^p(X)$ given by $[Z] \in \text{CH}^p(X) \mapsto [Z] \in \text{Hdg}^p(X)$.

Conjecture 2.2.7 (The Hodge Conjecture)

Let X be a smooth projective variety. Let $\alpha \in \text{Hdg}^1(X)$. Then there exists $N \in \mathbb{N}$ such that $N\alpha \in \text{im}(\text{cl})$.

2.3 The Lefschetz Theorems

Theorem 2.3.1 (Lefschetz Hyperplane Theorem)

Let X be a compact Kähler manifold. Let $V \subseteq X$ be a smooth variety of dimension $\dim(X) - 1$. Then the restriction map

$$H^k(X, \mathbb{Z}) \rightarrow H^k(V, \mathbb{Z})$$

induces isomorphisms for $0 \leq k < \dim(X) - 1$ and is injective for $k = \dim(X) - 1$.

Theorem 2.3.2 (The Lefschetz (1, 1) Theorem)

Let X be a compact Kähler manifold. Let $\alpha \in H^2(X, \mathbb{Z})$. Then α is the first Chern class of a line bundle if and only if $\alpha \in H^{1,1}(X)$.

Proposition 2.3.3

Let X be a compact Kähler manifold. Let $D \in \text{Div}(X)$ be a divisor. Then we have

$$c_1(\mathcal{L}(D)) = [D] \in H^2(X; \mathbb{Z})$$

Corollary 2.3.4

Let X be a compact Kähler manifold. Then the Hodge conjecture is true for $p = 1$.

2.4 Projective Curves

Proposition 2.4.1

Let X be a smooth projective curve over $\mathbb{C}$. Then X^{an} is homeomorphic to $\Sigma_{p_g(X)}$ the orientable surface of genus $p_g(X)$.

Corollary 2.4.2

There is an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces.

3 Mixed Hodge Structures

4 Periods and Period Maps